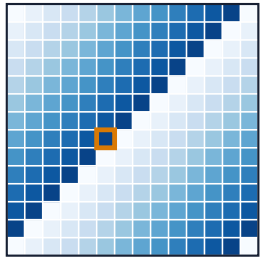
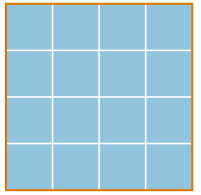
Does the finer ranking predict the internal response?

Compare the two explanations using the same target, image and intervention area.

a Choose the spatial unit



14 × 14 patches



ONE 16 × 16-pixel patch

16 cells of 4 × 4 pixels

Native scale

196 units · 16 × 16 pixels per unit

Fine scale

3,136 units · 4 × 4 pixels per unit

The same unit size is used for both methods.

b Rank fine cells

14	10	6	13
9	3	1	7
15	4	2	8
16	11	5	12

NNS

Can rank cells within a patch

Illustrative local order: 1 = first

1	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1

Token baseline

Same score within this patch

Example: cA = 16 → score 1 per cell

Actual ranking uses all 3,136 cells.

Ties: 5 paired permutations for both methods.

c Restore or remove

Insertion



0 ½ 1

Deletion



0 ½ 1

Better ranking: higher insertion AUC and lower deletion AUC.

21 fractions · evaluate the same internal f

Masks are enlarged schematics.

Average 5 repeats within each image → $\Delta FID_i = F_i(NNS) - F_i(token)$ → **Image median + 95% CI**

$F = \frac{1}{2}(AUC\ insertion - AUC\ deletion)$

Paired bootstrap uses images as independent units.

Illustrative maps, ranks and masks. Gray reference: normalized zero / RGB 0.5. Actual intervention fraction is used for AUC.